

Day 2 – Summary

→ Control Flow & Functions

1. Control Flow

- `if / elif / else` – Used to make decisions.
- `for` loop – Repeats code for each item.
- `while` loop – Repeats code while a condition is true.
- `break` – Stops a loop.
- `continue` – Skips the current iteration.
- `pass` – Does nothing; used as a placeholder.
- `range()` – Generates a sequence of numbers.
- List comprehension – Creates a list in a short way.

2. Functions

- Function – A reusable block of code.
- Parameters – Values passed into a function.
- Default arguments – Parameters with predefined values.
- `return` – Sends a result back from a function.
- `*args` – Accepts multiple positional arguments.
- `**kwargs` – Accepts multiple keyword arguments.
- Multiple return values – A function can return more than one value.
- Lambda – A small one-line anonymous function.

3. Scope and LEGB

- Local – Variable inside a function.
- Enclosing – Variable inside an outer function.
- Global – Variable outside functions.
- Built-in – Python's predefined names.

Python searches variables in this order:

Local → Enclosing → Global → Built-in

→ Data Structures + File Handling

4. Data Structures

Lists

- Ordered and changeable collection.
- Allows duplicate values.
- Supports CRUD operations.
- Important operations:
 - Indexing
 - Slicing
 - `append()`
 - `remove()`
 - `sort()`
 - `reverse()`

Tuples

- Ordered collection.
- Cannot be changed after creation.
- This property is called immutability.

Sets

- Stores unique values.
- Common operations:
 - `union()`
 - `intersection()`
 - `difference()`

Dictionaries

- Stores data as key-value pairs.
- Important methods:
 - `.get()`
 - `.keys()`
 - `.items()`
 - `.values()`

- Supports updating, deleting, and dictionary comprehensions.

5. Nested Data Structures

- Nested list: A list inside another list.
- Nested dictionary: A dictionary inside another dictionary.
- Useful for storing structured data such as student details and marks.

6. File Handling

JSON

- Used to store structured data.
- `json.dump()` – Writes Python data to a JSON file.
- `json.load()` – Reads data from a JSON file.

CSV

- Means Comma-Separated Values.
- Useful for table-like data.
- `csv.writer()` – Writes rows to a CSV file.
- `csv.reader()` – Reads rows from a CSV file.

7. Error Handling

- `try` – Contains code that may cause an error.
- `except` – Handles the error safely.

Common errors practiced:

- `ValueError`
- `ZeroDivisionError`
- `FileNotFoundError`

8. Final Project – Contact Book

The Contact Book uses:

- Dictionary to store contacts
- Functions to add, view, search, and delete contacts
- JSON file to save contacts permanently

- `try/except` to handle a missing JSON file

DAY 2 Check List

- Control flow
- Functions
- Scope and LEGB
- Lists
- Tuples
- Sets
- Dictionaries
- Nested structures
- JSON
- CSV
- Error handling
- Contact Book with JSON persistence